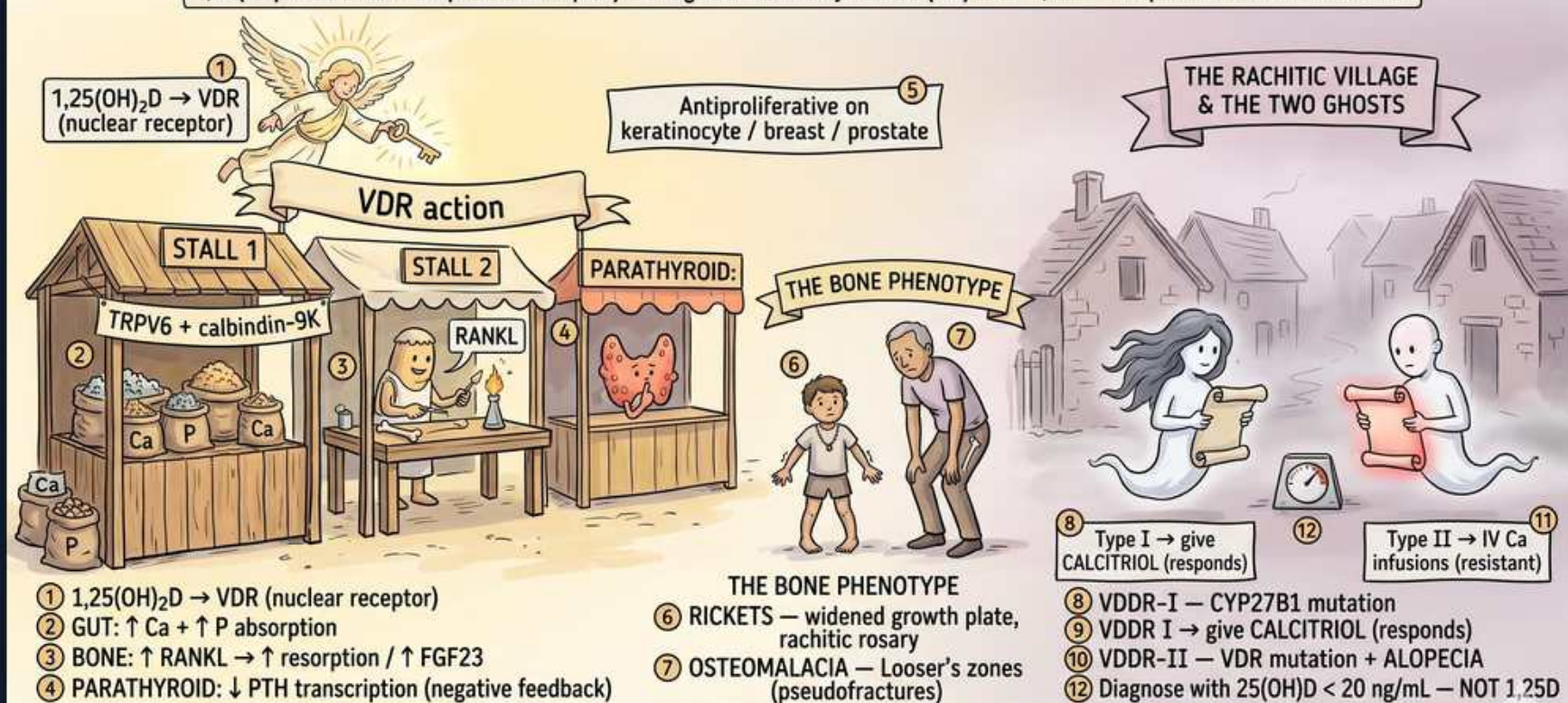


Parathyroid/Bone — Vitamin D Action & Deficiency (Marketplace)

THE VITAMIN D MARKETPLACE & THE RACHITIC VILLAGE — VDR ACTION & DEFICIENCY

1,25(OH)₂D acts via VDR (nuclear receptor) — diagnose deficiency with 25(OH)D not 1,25D — alopecia means VDR mutation



VDR is in gut, bone, parathyroid. Order 25(OH)D — 1,25D often normal in deficiency due to PTH compensation.
Alopecia means VDR mutation (type II) — calcitriol does NOT work, only IV Ca. CKD: ↑ FGF23 + retained P → low 1,25D.

1. 1,25-D acts via VDR

WHAT YOU SEE

Banner: 1,25(OH)₂D acts via VDR (nuclear receptor); diagnose deficiency with 25(OH)D not 1,25-D; alopecia means VDR mutation

WHAT IT ENCODES

Vitamin D activation pathway: skin (7-dehydrocholesterol + UVB) and diet provide vitamin D₃/D₂ -> LIVER 25-hydroxylase makes 25(OH)D (calcidiol, the STORAGE form and the one you measure) -> KIDNEY 1-alpha-hydroxylase (CYP27B1, stimulated by PTH/low PO₄, inhibited by FGF23/high Ca) makes 1,25(OH)₂D (CALCITRIOL, the ACTIVE form). Calcitriol acts through the nuclear VDR (a transcription factor) to drive intestinal Ca/PO₄ absorption.

BOARD TRAP

WRONG: ordering a 1,25-D (calcitriol) level to diagnose vitamin D DEFICIENCY. 1,25-D can be NORMAL or even HIGH in deficiency because secondary hyperPTH upregulates 1-alpha-hydroxylase. Deficiency is defined by a low 25(OH)D — that's the storage marker. Reserve 1,25-D for specific cases (CKD, hereditary rickets, granulomatous hypercalcemia).

2. Gut: Ca + PO₄ absorption

WHAT YOU SEE

Stall 1 (gut): TRPV6 + calbindin-9K; up Ca and up PO₄ absorption

WHAT IT ENCODES

The PRIMARY action of 1,25-D is to increase INTESTINAL absorption of both calcium (via TRPV6 channel and calbindin-9K) and PHOSPHATE. This is the dominant route by which active vitamin D raises serum Ca and PO₄. Without it, gut Ca absorption falls to a fraction of normal.

BOARD TRAP

WRONG: assuming vitamin D, like PTH, lowers phosphate. Vitamin D raises BOTH Ca AND phosphate (gut absorption), whereas PTH raises Ca but LOWERS phosphate (renal phosphaturia). This is why vitamin D TOXICITY produces high Ca AND high phosphate, while primary hyperPTH gives high Ca with LOW phosphate.

3. Bone: RANKL + FGF23

WHAT YOU SEE

Stall: BONE — up RANKL -> up resorption; up FGF23

WHAT IT ENCODES

In bone, 1,25-D promotes osteoblast expression of RANKL, which activates osteoclasts -> bone resorption that mobilizes Ca/PO₄ (important at high concentrations / when gut supply is short). 1,25-D also stimulates osteocyte FGF23, providing negative feedback (FGF23 then suppresses 1-alpha-hydroxylase and promotes phosphaturia).

BOARD TRAP

WRONG: thinking vitamin D only BUILDS bone. Adequate vitamin D is needed for mineralization, but 1,25-D's direct bone action is actually pro-RESORPTIVE (via RANKL) to defend serum Ca; the bone-building benefit comes indirectly from supplying Ca/PO₄ substrate for mineralization. Deficiency, not excess, causes the under-mineralized soft bone (rickets/osteomalacia).

4. Parathyroid feedback

WHAT YOU SEE

Parathyroid stall: down PTH transcription (negative feedback)

WHAT IT ENCODES

1,25-D directly SUPPRESSES PTH gene transcription in the parathyroid (negative feedback), complementing the CaSR-mediated suppression by calcium. This is the basis for using active vitamin D analogs (calcitriol, paricalcitol) to control secondary hyperparathyroidism in CKD.

BOARD TRAP

WRONG: forgetting that vitamin D deficiency is a major DRIVER of secondary hyperPTH. Low 1,25-D removes this brake AND lowers serum Ca, both of which raise PTH. So vitamin D deficiency classically shows HIGH PTH — a 'normal' PTH in a deficient patient may itself be inappropriate.

5. Rickets (open plates)

WHAT YOU SEE

Bone Phenotype #6: RICKETS — widened growth plate, rachitic rosary (children, open growth plates)

WHAT IT ENCODES

RICKETS is defective mineralization at the OPEN growth plates of CHILDREN: widened/frayed metaphyses, bowing of long legs (genu varum/valgum), rachitic rosary (costochondral beading), frontal bossing, delayed fontanelle closure. Labs: low/normal Ca, low PO₄, high ALP, high PTH, low 25(OH)D.

BOARD TRAP

WRONG: using 'rickets' and 'osteomalacia' interchangeably. RICKETS = the CHILD with OPEN growth plates (growth-plate cartilage doesn't mineralize). OSTEOMALACIA = the same biochemistry in the ADULT after growth plates have fused (defective mineralization of mature osteoid). Age/growth-plate status is the discriminator.

6. Osteomalacia (Looser zones)

WHAT YOU SEE

Bone Phenotype #7: OSTEOMALACIA — Looser's zones (pseudofractures) in adults

WHAT IT ENCODES

OSTEOMALACIA is impaired mineralization of mature adult bone -> diffuse BONE PAIN, PROXIMAL MUSCLE WEAKNESS/myopathy (waddling gait), and LOOSER'S ZONES (pseudofractures — radiolucent bands perpendicular to the cortex, classically at the femoral neck, pubic rami, scapula). Labs: low/normal Ca, low PO₄, HIGH ALP, high PTH, low 25(OH)D.

BOARD TRAP

WRONG: calling diffuse bone pain with low Ca/low PO₄/HIGH ALP 'osteoporosis.' Osteoporosis has NORMAL Ca, PO₄, and ALP and is a quantity (low bone mass) problem; osteomalacia is a QUALITY (mineralization) problem with abnormal labs and Looser zones. The chemistry separates them.

7. Antiproliferative role

WHAT YOU SEE

Banner #5: antiproliferative on keratinocyte / breast / prostate

WHAT IT ENCODES

Beyond mineral metabolism, the VDR mediates ANTIPROLIFERATIVE and pro-differentiation effects on keratinocytes, breast, prostate, and immune cells. This underlies the use of vitamin D analogs (e.g., calcipotriol) topically for PSORIASIS.

BOARD TRAP

WRONG: extrapolating these effects into claims that vitamin D supplements prevent cancer. The antiproliferative biology is real at the cellular/VDR level, but board answers should not equate it with proven clinical cancer prevention — the classic exam role of vitamin D remains mineral/bone homeostasis (and topical analogs for psoriasis).

8. Type I rickets (1-alpha def)

WHAT YOU SEE

Rachitic Village ghost #8 / #10: Type I — VDDR-I, CYP27B1 mutation, give CALCITRIOL (responds)

WHAT IT ENCODES

Vitamin-D-DEPENDENT rickets type I (VDDR-I): loss-of-function of 1-alpha-hydroxylase (CYP27B1) -> the kidney can make 25(OH)D but CANNOT convert it to active 1,25-D. Labs: NORMAL/high 25(OH)D but LOW 1,25-D, low Ca, low PO₄, high PTH. Treatment is CALCITRIOL (active vitamin D), which bypasses the missing enzyme — patients respond well because the VDR is intact.

BOARD TRAP

WRONG: treating VDDR-I with ordinary (inactive) vitamin D₂/D₃. The enzyme to activate it is missing, so plain cholecalciferol won't work — give the ACTIVE form, CALCITRIOL. The intact receptor (no alopecia) is the clue that activated vitamin D will be effective.

9. Type II rickets (VDR) + alopecia

WHAT YOU SEE

Rachitic Village ghost #9 / #11: Type II — VDR mutation + ALOPECIA, resistant, give IV Ca infusions

WHAT IT ENCODES

Vitamin-D-dependent rickets type II (VDDR-II / hereditary vitamin-D-resistant rickets): a VDR mutation causes END-ORGAN RESISTANCE to 1,25-D. Labs: VERY HIGH 1,25-D (compensatory), low Ca, low PO₄, high PTH. Hallmark physical clue: ALOPECIA (often total). Calcitriol does NOT work (receptor is broken) — treat with high-dose calcium (often IV calcium infusions) and high-dose calcitriol attempts.

BOARD TRAP

WRONG: giving calcitriol to a VDDR-II child and expecting a response. The VDR itself is defective, so even sky-high 1,25-D can't act — therapy is high-dose/IV CALCIUM to bypass the gut absorption defect. ALOPECIA is the giveaway pointing to a VDR mutation (type II) rather than the enzyme defect (type I).

10. Diagnose with 25(OH)D <20

WHAT YOU SEE

Banner #12: diagnose deficiency with 25(OH)D <20 ng/mL — NOT 1,25-D

WHAT IT ENCODES

Vitamin D status is defined by serum 25(OH)D (the storage form): DEFICIENT <20 ng/mL, INSUFFICIENT 20-30 ng/mL, sufficient >30 ng/mL. Treat deficiency with ergocalciferol/cholecalciferol ~50,000 IU weekly for 6-8 weeks, then maintenance 800-2000 IU/day, plus calcium.

BOARD TRAP

WRONG: using 1,25-D (calcitriol) to screen for or confirm deficiency. 1,25-D is tightly regulated and frequently normal/high even when stores are depleted (secondary hyperPTH drives 1-alpha-hydroxylase). Order 25(OH)D for diagnosis; reserve 1,25-D for CKD, granulomatous hypercalcemia, or suspected hereditary rickets.

11. CKD: low 1,25-D, high FGF23

WHAT YOU SEE

Bottom banner: CKD -> FGF23 up + retained P -> low 1,25-D

WHAT IT ENCODES

In CKD the kidney loses functional 1-alpha-hydroxylase AND phosphate retention raises FGF23, both of which suppress 1,25-D production -> low ACTIVE vitamin D -> reduced gut Ca absorption -> hypocalcemia -> secondary (then tertiary) hyperPTH. The picture is high phosphate, low/low-normal Ca, low 1,25-D, high PTH (CKD-MBD).

BOARD TRAP

WRONG: treating CKD bone-mineral disease with plain ergocalciferol/cholecalciferol alone. The diseased kidney can't activate it — give ACTIVE vitamin D (calcitriol or paricalcitol) PLUS phosphate binders (and cinacalcet for high PTH). Plain vitamin D only repletes 25-D stores; it won't fix the activation block.

12. Vit D negative feedback (gut)

WHAT YOU SEE

Bottom banner: vit D is gut, bone, parathyroid; order 25(OH)D; 1,25-D often normal even in deficiency due to PTH compensation

WHAT IT ENCODES

Tie-together: vitamin D acts on GUT (main: absorb Ca/PO₄), BONE (RANKL resorption + supply substrate for mineralization), and PARATHYROID (suppress PTH transcription). In deficiency, secondary hyperPTH upregulates 1-alpha-hydroxylase so 1,25-D may stay 'normal' despite depleted 25-D stores — order 25(OH)D, not 1,25-D, to diagnose.

BOARD TRAP

WRONG: concluding 'a normal 1,25-D rules out vitamin D deficiency.' Because PTH-driven compensation keeps 1,25-D normal (or high) even as 25-D falls, a normal calcitriol does NOT exclude deficiency. The storage form 25(OH)D is the only reliable diagnostic measure.